

Stanisław Słowiński

TELECOMMUNICATIONS STUDENT



Motivated engineering student with a passion for astronomy, space instrumentation and writing the software to bring hardware to life.

Project Portfolio: <https://jianskycontinuum.github.io/>

Education

2022 – present
Graduating
June 2027

Warsaw University of Technology (WUT)
Bachelor of Engineering in Telecommunications (6th semester)

Latest honors and awards

November 2025

WUT's Headmaster Scholarship for top 8% students based on academic excellence.

A scholarship aimed at the best students based on their performance from two previous semesters.
This is the second time I have received this distinction.

June 2025

2025 IEEE AP-S Student Design Contest finalist

As team leader, I led my group to the finals of the 2025 IEEE AP-S Student Design Contest; we presented our project at the IEEE AP-S/URSI North American Radio Science Meeting in Ottawa, Canada in July 2025.

Personal projects

October 2024 –
June 2025

Time Modulated Antenna Array with beamforming capabilities and real-time tracking

Aside from formal duties of a leader (organization, writing reports etc.) I was responsible for many key technical aspects of the project

My roles \ developed skills:

- **Performed hardware assembly and soldering (SMD/THT)**, building a robust physical prototype and utilizing various laboratory equipment for testing.
- **Documented system design and progress by writing detailed reports** and step-by-step instructions to allow reproduction of the system as per the contest's guidelines
- **Programmed microcontrollers and integrated wireless modules (Zigbee)**
- **Designed 3D models** for mechanical components of the setup

- **Designed and implemented a real-time tracking algorithm**, ensuring high positioning accuracy and system reliability.

Adaptive antenna with beamforming capabilities for tracking weather satellites on the low-earth orbit

September 2025 – present
 Focused on developing a prototype ground station, integrating custom electronics (PCB design in Altium) with software control systems. My roles \ developed skills:

- **Conducted functional testing, troubleshooting and debugging of the electro-mechanical setup** using laboratory equipment to ensure reliable tracking mechanics.
- **Integrated and controlled** a satellite tracking rotor using Python within a Linux environment, managing accurate angular positioning.

Student initiatives

March 2025 - present
Founder and President - Student Club of Antenna Design
 I established a student club focused on antenna design and applied electromagnetics. Until June 2025 the main focus was the mentioned IEEE contest, however I'm now working on a new project – an antenna array capable of tracking weather satellites on the low Earth orbit (LEO) through the use of electronic beamforming

Community service and volunteering

2023 - present
Dzieło Na Misji Volunteer
 In the past summer of 2024 I taught English and mathematics in a local high school and primary school in rural Tanzania for two months as a volunteer of the Dzieło Na Misji Foundation.

Languages

Polish	Native
English	C1/C2, full professional proficiency

Computer skills

Programming	Python, Java, MATLAB, C, C++
Software	AutoCAD, Fusion 360, FEKO, Altium Designer, LTspice

Publications

S. Słowiński, M. Bandyach, G. Bogdan, and Y. Yashchyshyn, "Sixteen-Element Time-Modulated Antenna Array With Beam Steering in ISM Band," presented at KRI7, Gdańsk 2025.